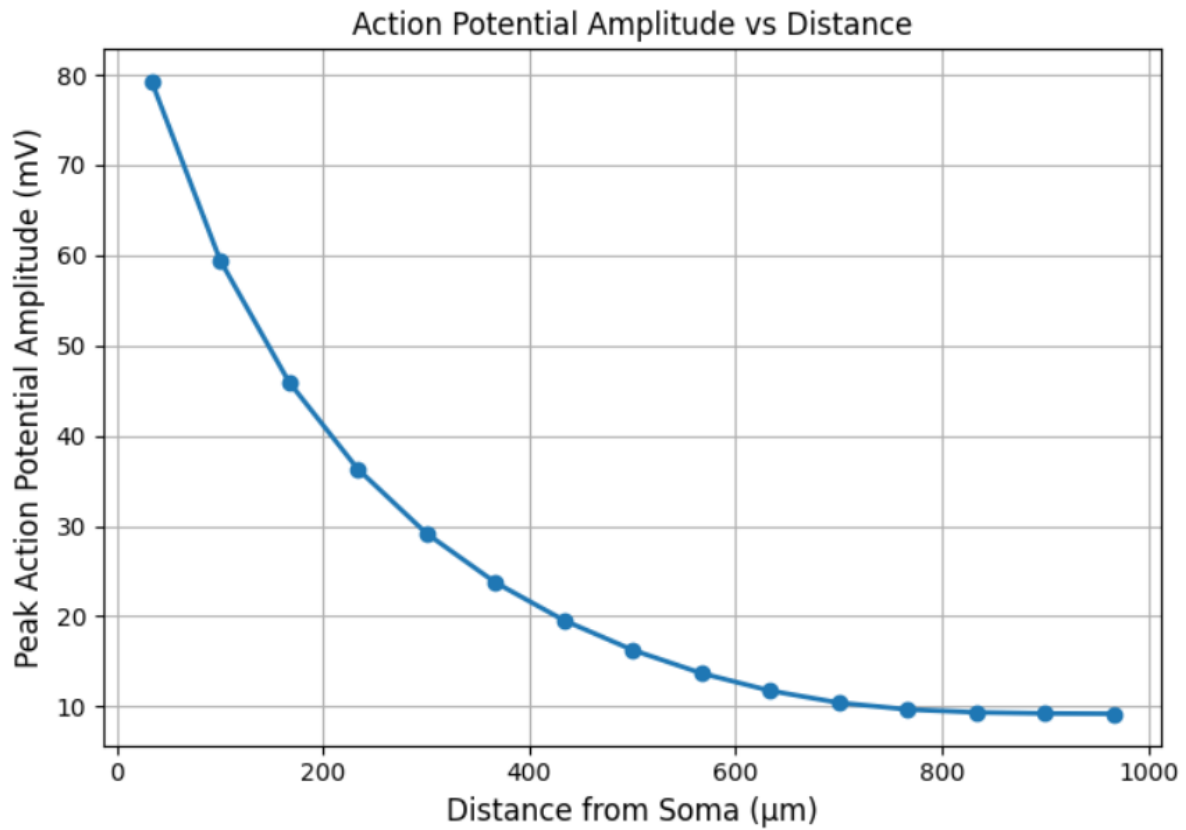
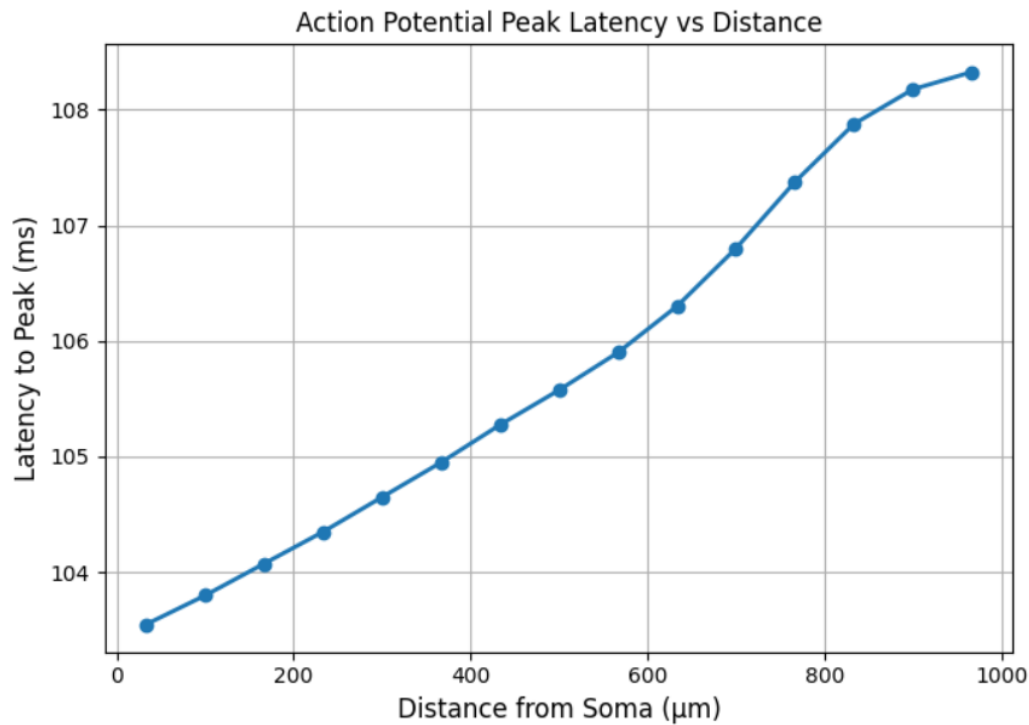


3.

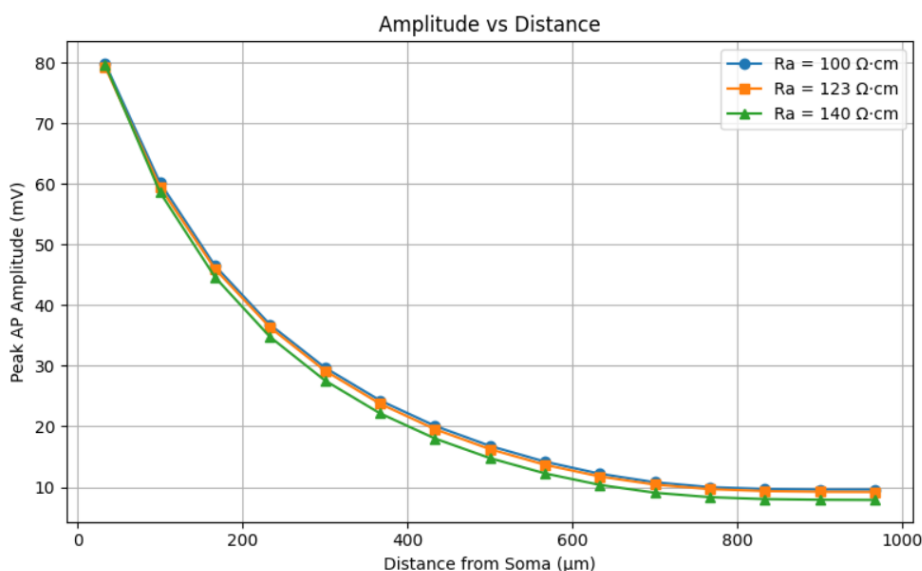


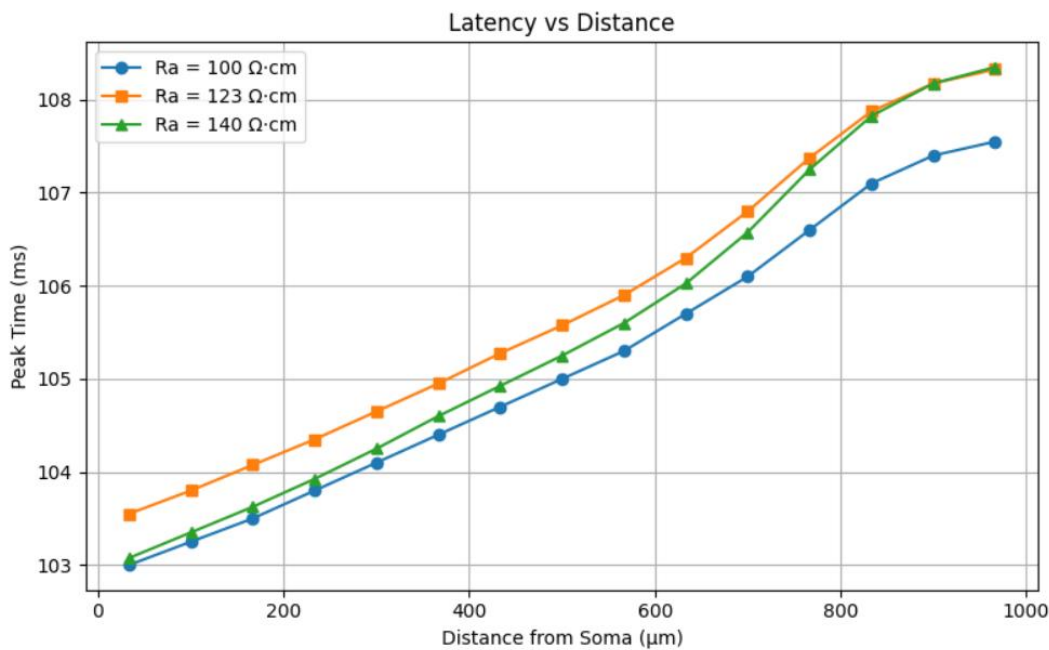
Interpretation: As the distance from soma increases, the action potential peak amplitude decreases.



Interpretation: As the distance from the soma increases, the latency to reach the action potential peak increases. This is because the action potential propagates along the dendrite with a finite speed, so regions farther from the soma receive the signal later.

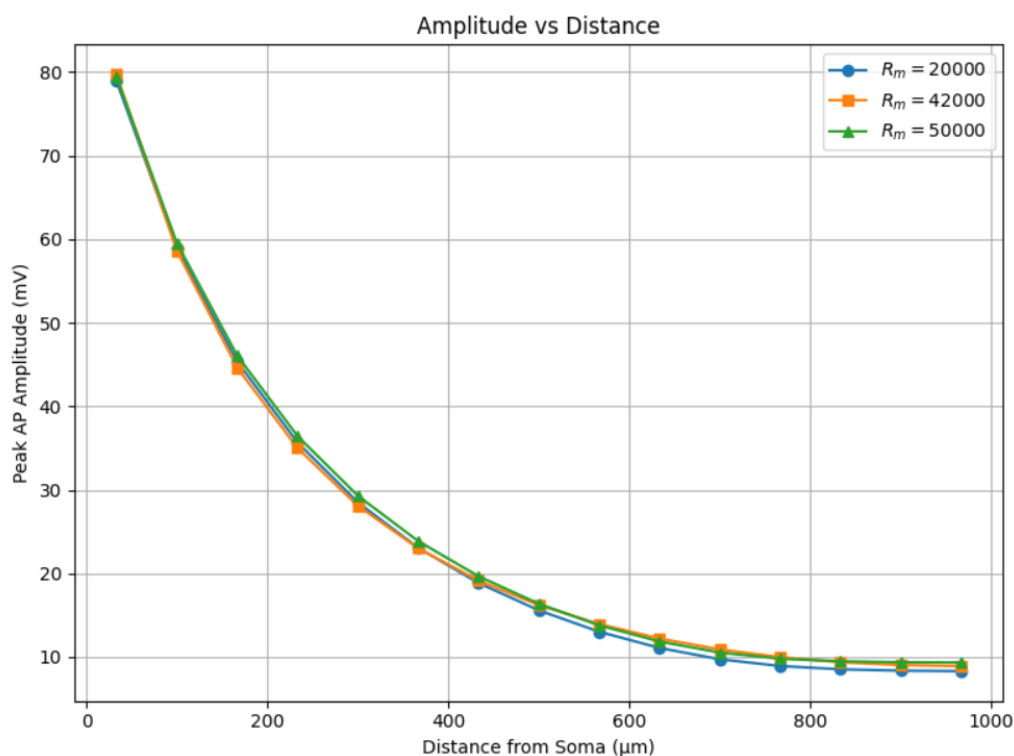
As a function of R_a

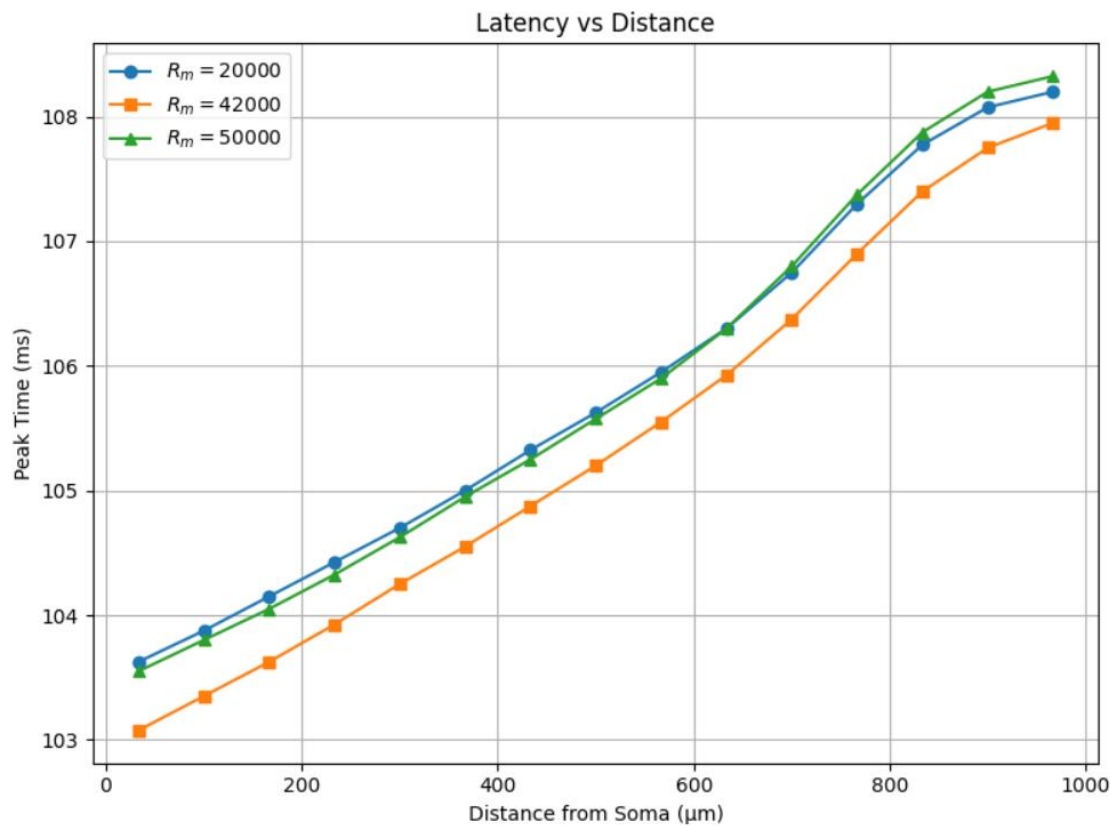




Interpretation: Increasing the axial resistance slows the spread of current along the dendrite. So, at the same distance amplitude of the action potential decreases and the latency to reach the action potential peak increases because the action potential takes time to propagate along the dendrite

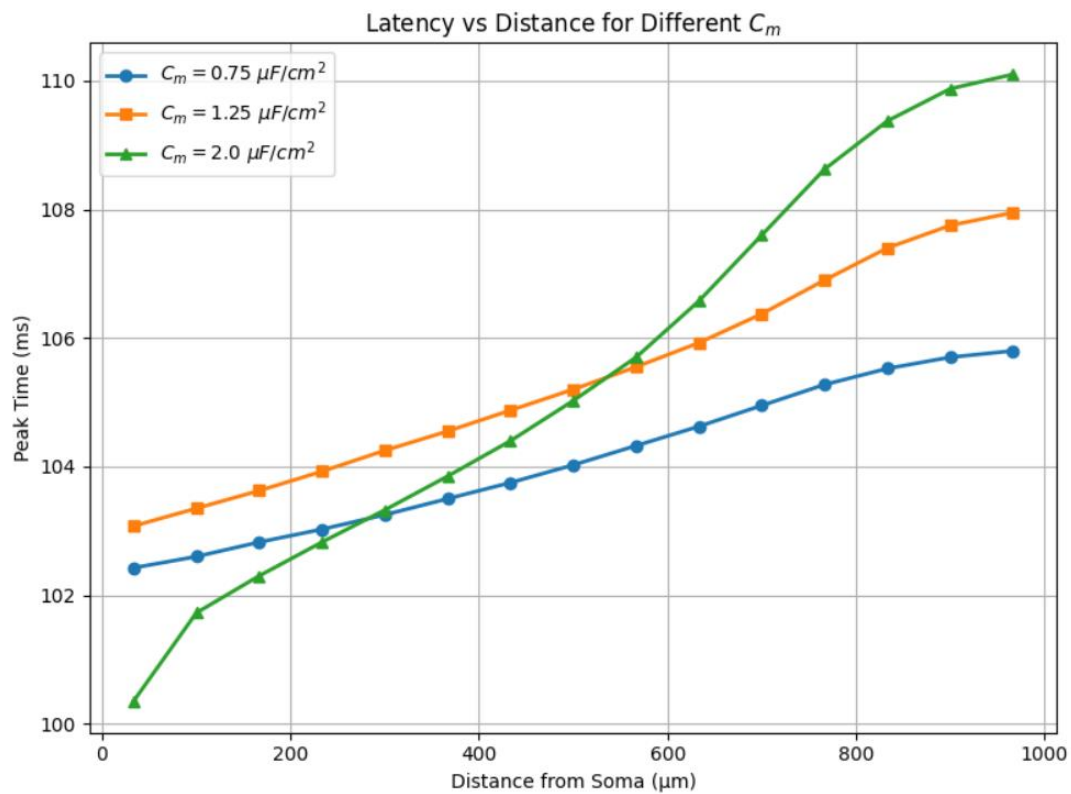
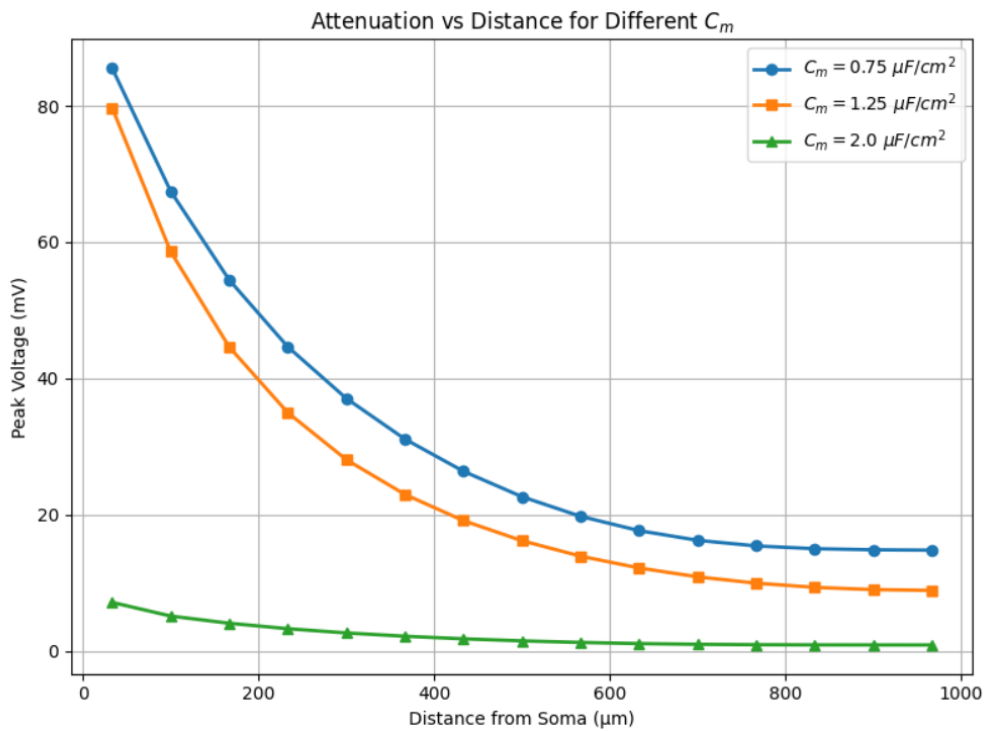
As a function of R_m





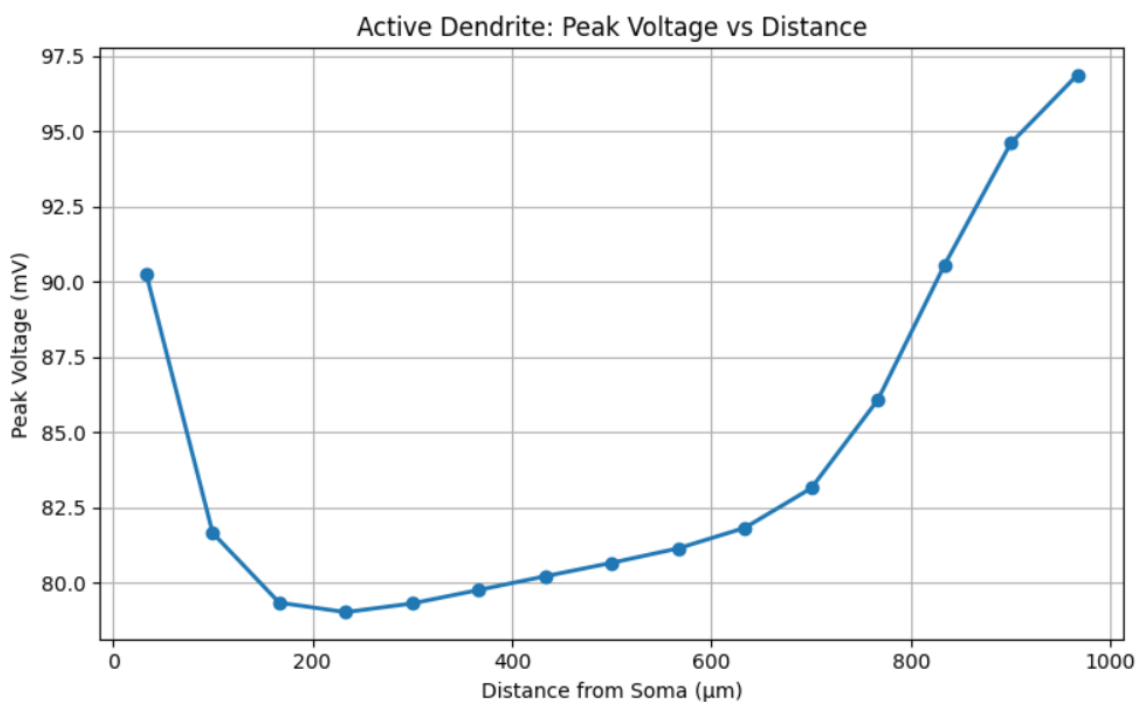
Interpretation: Increasing the membrane resistance (R_m) reduces the leakage of current through the membrane. As a result, the voltage attenuates less as it travels along the dendrite, and the action potential amplitude remains larger at greater distances from the soma. Since less current is lost through the membrane, the depolarization spreads more efficiently. The propagation delay may decrease slightly, although the main effect of increasing R_m is the reduction in signal attenuation rather than a large change in latency.

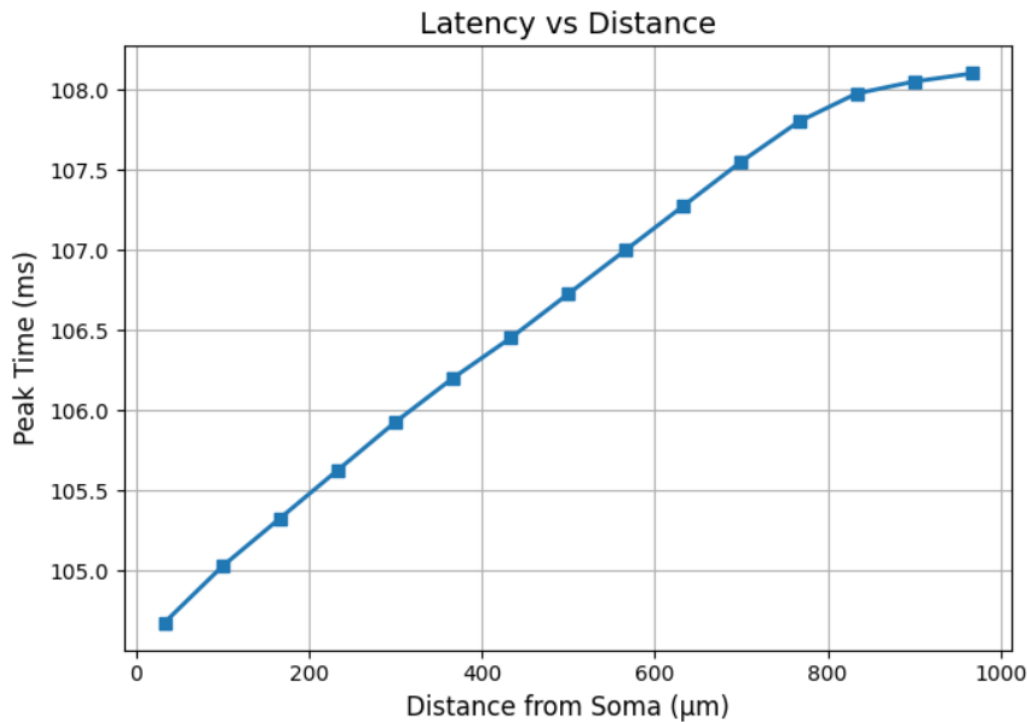
As a function of C_m



Interpretation: Increasing the membrane capacitance (C_m) causes the membrane to charge more slowly because more charge is required to change the membrane potential. As a result, the action potential propagates more slowly, leading to an increase in latency with distance. In addition, the signal undergoes greater attenuation, so the peak voltage decreases more rapidly along the dendrite. Conversely, a lower C_m allows the membrane to charge more quickly, resulting in faster propagation and less attenuation.

Active dendrites

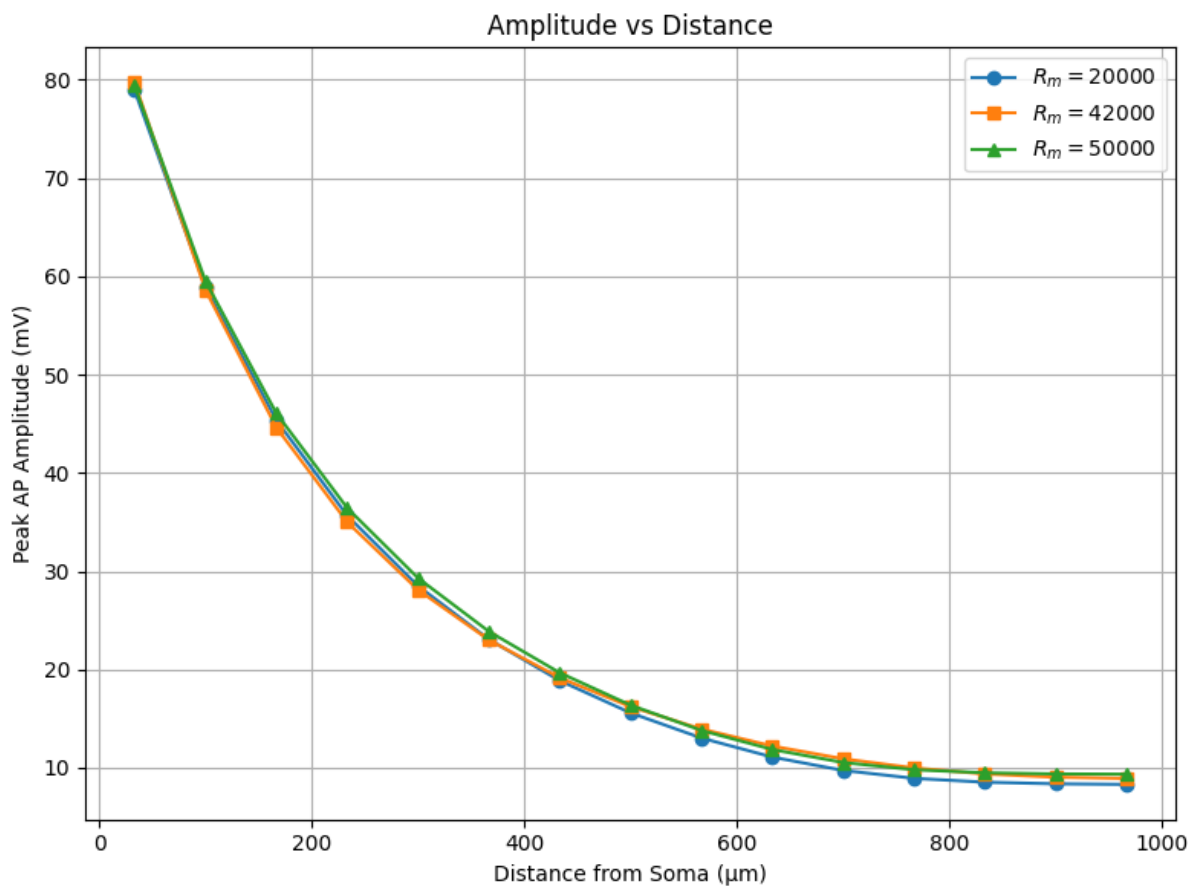


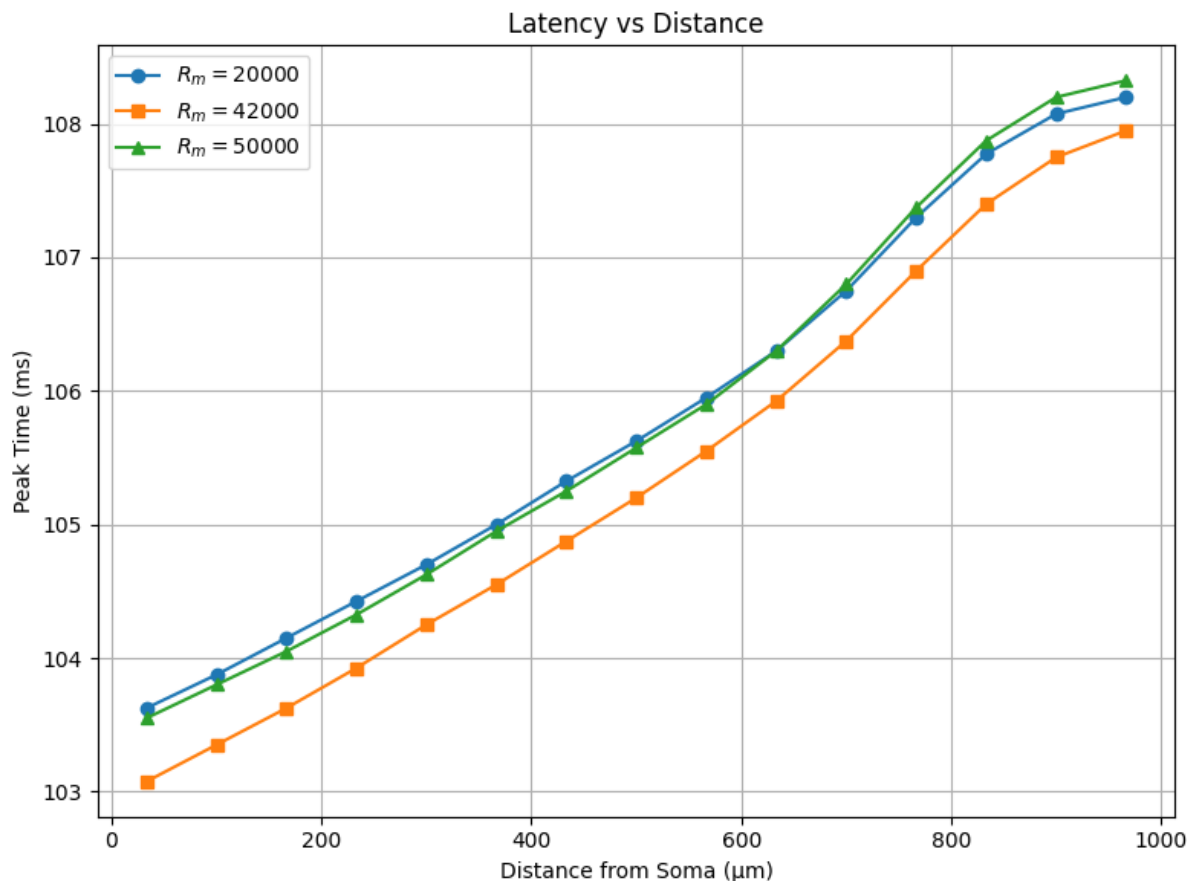


Interpretation: As the distance from the soma increases, the latency to reach the action potential peak also increases because the action potential still requires time to propagate along the dendrite. However, unlike the passive dendrite, the peak voltage does not continuously decrease with distance. After a small initial drop, the amplitude is maintained and even increases toward the distal end because the voltage-gated sodium channels in the dendrite actively regenerate the action potential. Thus, active backpropagation greatly reduces signal attenuation and allows action potentials to

propagate more reliably over long distances
compared with passive propagation.

As a function of R_m

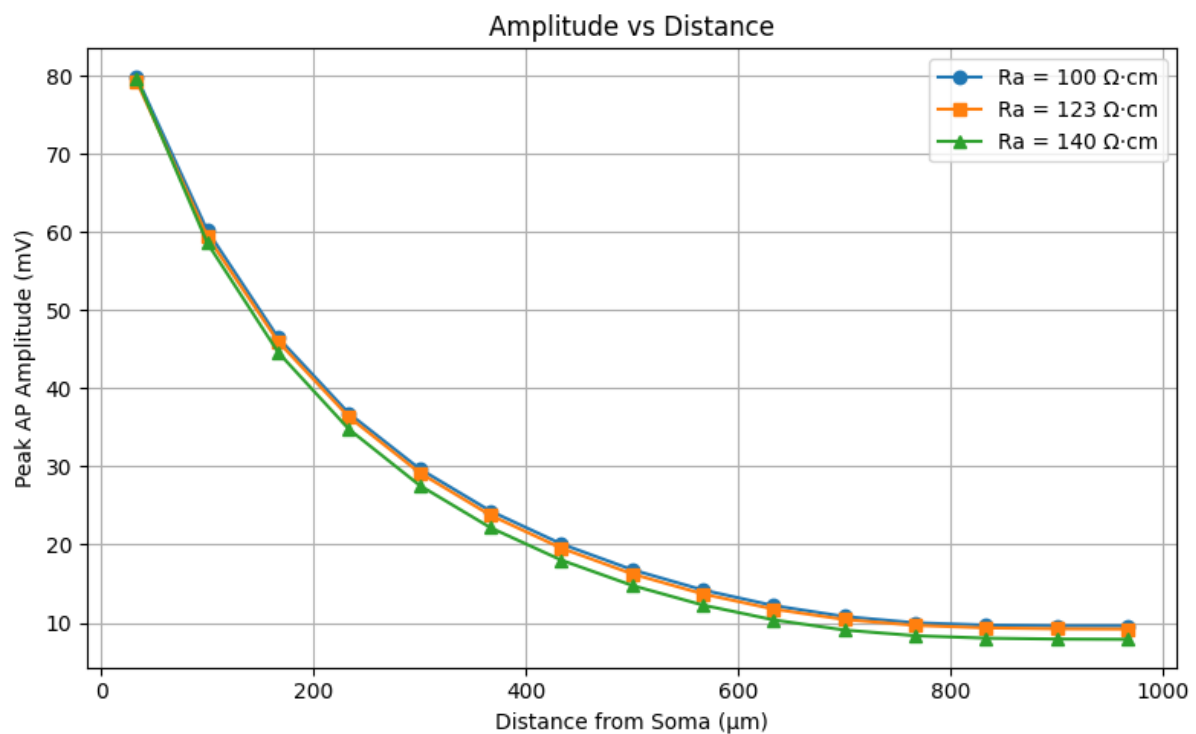


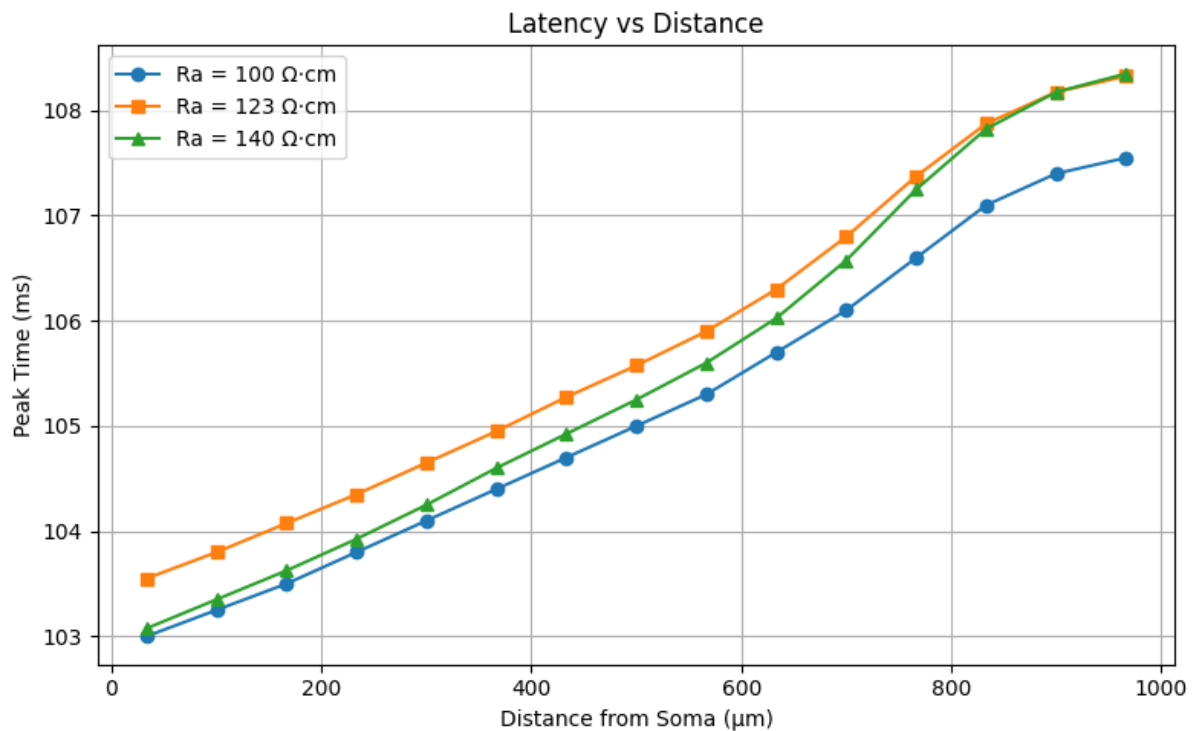


Interpretation: In the active dendrite, the action potential is continuously regenerated by the voltage-gated sodium channels as it propagates. Therefore, changing R_m has only a modest effect on both the propagation delay and the action potential amplitude. A higher R_m slightly reduces membrane current leakage, allowing the action potential to maintain a marginally larger amplitude, while the latency changes only slightly

because propagation is dominated by the active regeneration of spikes rather than passive current spread. Thus, compared with the passive dendrite, the influence of R_m is much smaller in the active case.

As a function of R_a

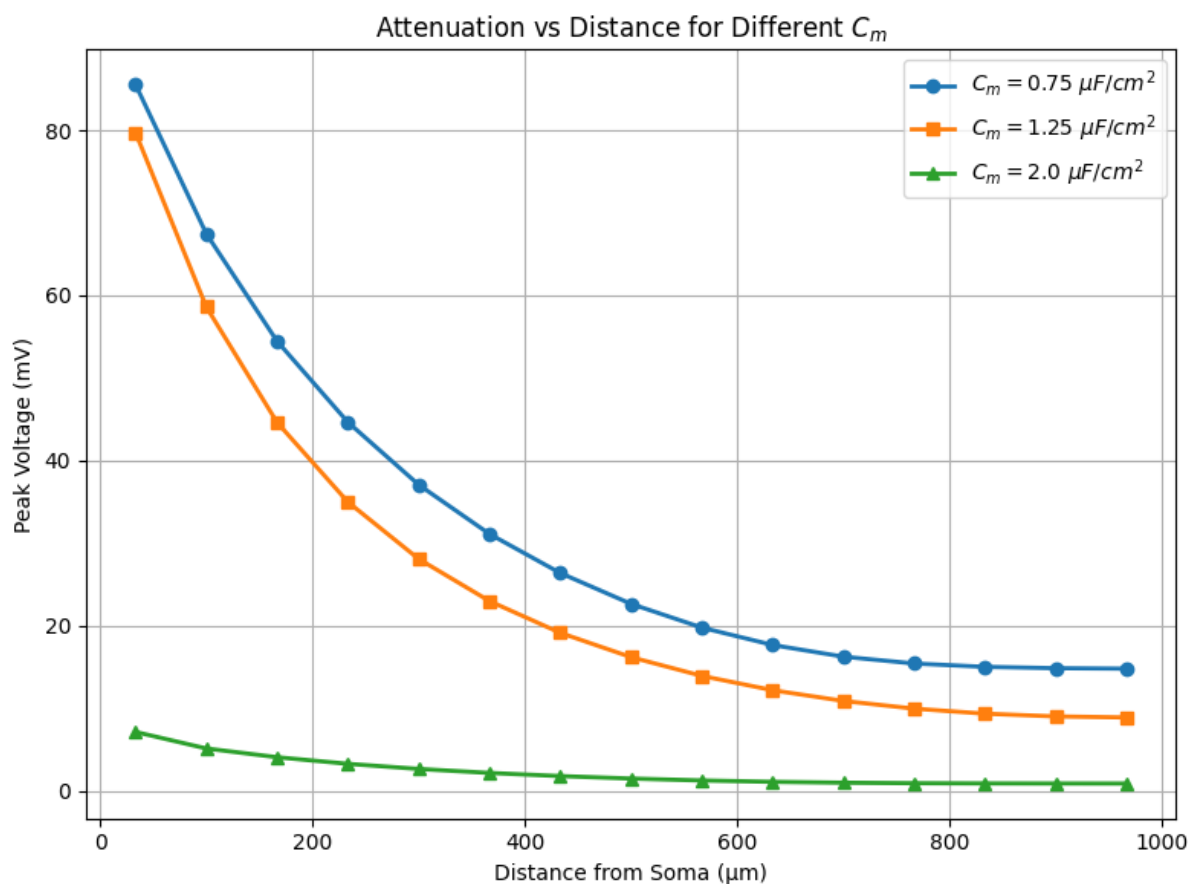


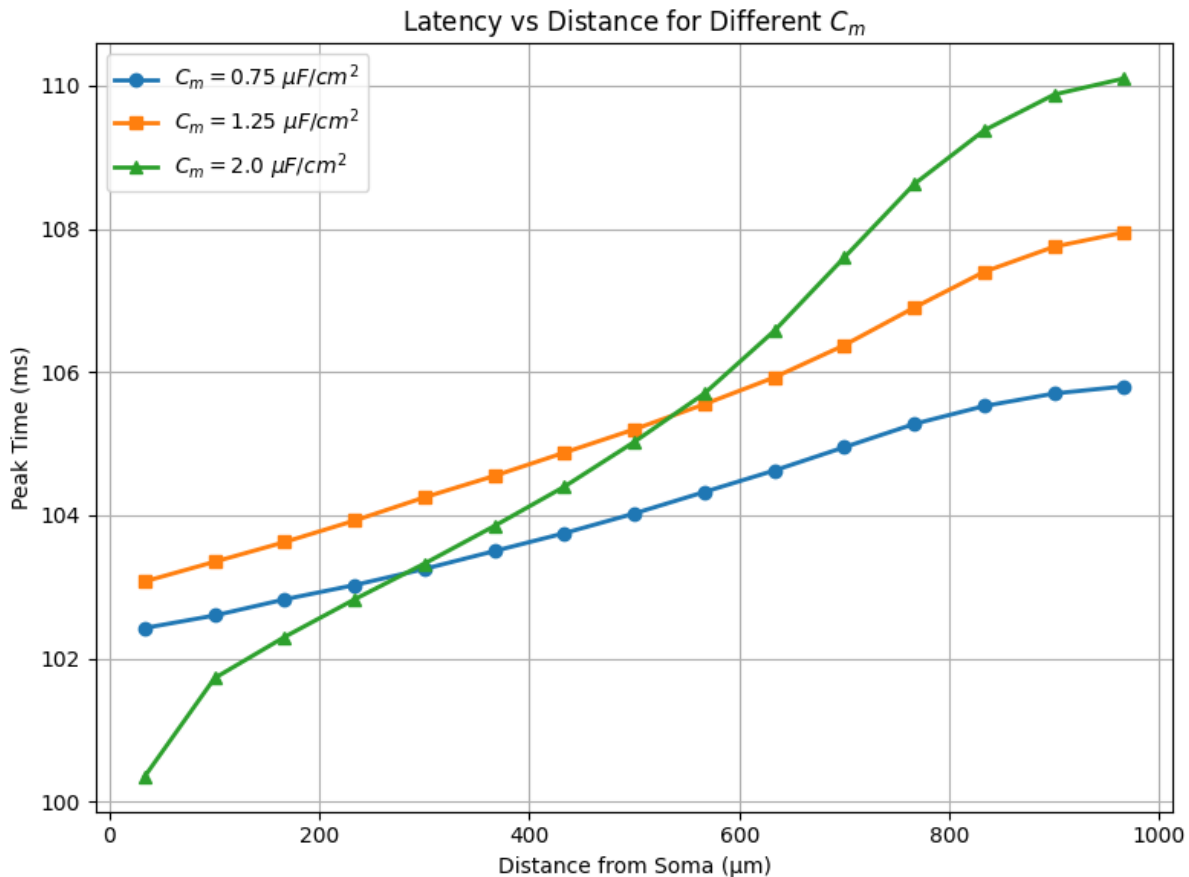


Interpretation: As the value of axial resistance R_a increases, the charge will face more resistance to propagate along the length of the dendrite. Hence, the peak action potential amplitude will be less at the distance as we keep increasing the R_a value. However, since the dendrites are active and can regenerate the action potential themselves the difference is quite small. Again, the latency or the time required to fire action potential at the same distance from the soma also decrease as we

decrease R_a . However, the differences are not so significant as the dendrites are active and can regenerate the action potential.

As a function of C_m





Interpretation: As the membrane capacitance (C_m) increases, each part of the dendrite needs more charge before it can depolarize. Because of this, the action potential takes longer to travel along the dendrite, so the peak time (latency) increases with both distance and C_m . The slower depolarization also delays the activation of sodium channels, making the regeneration of the action potential less effective. As a result, the peak voltage decreases more with distance when

C_m is high. When C_m is low, the membrane charges faster, so the action potential travels more quickly and maintains a higher peak voltage. Thus, even though both the soma and dendrites have Hodgkin–Huxley channels, increasing C_m slows propagation and reduces the spike amplitude along the dendrite. The $C_m = 1.25 \mu\text{F}/\text{cm}^2$ (green) curve shows a slightly lower latency than the $C_m = 0.75 \mu\text{F}/\text{cm}^2$ curve at some distances, which is an unexpected observation during simulation.